CAPSTONE PROJECT FINAL REPORT

# WorkHub

## Team Workflow & Event Operations Management System

A Full-Stack Collaborative Enterprise Platform for Mega-Events, Cross-Functional Departments, and Operational Portfolios


| | |
|---|---|
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| INSTITUTION | Vignan University – Engineering Faculty |
| CASE STUDY | Vignan Mahotsav 2027 Mega-Fest Operations |
| SUBMISSION | November 2026 • Version 1.0.0 (Production Release) |


**CORE ARCHITECTURAL HIGHLIGHTS**

- **Strict MERN stack:** MongoDB (Mongoose 8.x), Express.js, React 18 (Vite 6) and Node.js v22.
- **Deterministic engineering:** traditional software logic with no AI dependencies or black-box APIs.
- **4-tier operational hierarchy:** Organization → Workspace → Team / Department → Collaborative Room.
- **3 onboarding workflows:** shareable invite links, unique Member IDs (WH-XXXXXX) and directory search.
- **Quality assurance:** 20 / 20 automated integration tests passing and 0 Vite build warnings.

# CERTIFICATE OF ORIGINALITY

This is to certify that the capstone project entitled **“WorkHub – Team Workflow & Event Operations Management System”**, submitted by **Namitha Koduru** in partial fulfilment of the academic requirements, is a bona fide record of independent and original full-stack software development carried out under rigorous engineering methodologies.

All modules, data models, API endpoints and client interfaces documented herein represent actual implemented source code, without reliance on synthetic mock data or external artificial-intelligence generators.

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# Table of Contents

**1. Abstract . . . 4**

**2. Introduction & Background . . . 4**

2.1 Case study: Vignan Mahotsav 2027 . . . 4

**3. Problem Statement & Industrial Need . . . 5**

**4. Existing Systems vs. Proposed WorkHub System . . . 5**

**5. Project Objectives & Operational Scope . . . 5**

**6. Functional & Non-Functional Requirements . . . 6**

6.1 Functional requirements . . . 6

6.2 Non-functional requirements . . . 6

**7. Technology Stack & Architectural Rationale . . . 7**

**8. High-Level System Architecture . . . 8**

**9. Use Case Modeling & Actor Specifications . . . 8**

**10. Entity-Relationship Modeling & Constraints . . . 9**

**11. Database Design & Mongoose Schemas . . . 9**

**12. Application Module Breakdown . . . 10**

**13. Implementation Details & Technical Innovations . . . 10**

13.1 Unique Member ID generation hook . . . 10

13.2 Three-way onboarding controller logic . . . 10

13.3 Real-time synchronisation via Socket.IO . . . 11

**14. Verification, Testing & Test Case Execution . . . 11**

**15. Application Visual Walkthrough & Screenshots . . . 13**

**16. Results & Operational Analysis . . . 19**

**17. Conclusion & Future Roadmap . . . 19**

Future enhancements . . . 19

**18. References & Technical Citations . . . 19**

# 1. Abstract

Coordinating multi-faceted, high-stakes initiatives — annual collegiate festivals such as Vignan Mahotsav 2027 (20,000+ delegates), academic symposia and enterprise projects — is a severe operational bottleneck. Personal productivity tools offer no hierarchical synchronisation, which leads to fragmented communication, redundant effort, untracked dependencies and delayed task sign-offs.

This capstone presents **WorkHub**, an enterprise-grade team workflow and operations management system built on the MERN stack (MongoDB, Express.js, React 18 with Vite, and Node.js with Socket.IO). It introduces a four-tier hierarchy — **Organization → Workspace → Team / Department → Collaborative Room**. Each room is an isolated execution unit with its own Kanban board, multi-assignee task distribution, milestone tracking and audit log.

Three onboarding mechanisms make joining frictionless yet controlled: (1) shareable invite links, (2) direct addition by Unique Member ID (WH-XXXXXX) and (3) an organisational discovery directory with manager approval gates. A five-tier Role-Based Access Control (RBAC) engine enforces least-privilege governance.

The system is deliberately free of AI components and delivers deterministic accountability, a master event calendar, multi-collaborator review pipelines and a consolidated daily-standup blocker view. All 20 automated integration tests pass.

**Keywords:** MERN stack, workflow management, event operations, RBAC, Kanban, Socket.IO, real-time collaboration.

# 2. Introduction & Background

Large-scale project planning involves complex interdependencies between physical infrastructure, financial sponsorship, creative media, technical engineering and public relations. In universities and corporate departments alike, hundreds of people across specialised committees must be synchronised.

## 2.1 Case study: Vignan Mahotsav 2027

WorkHub was designed and seeded around the end-to-end execution of Vignan Mahotsav 2027, a national cultural and technical festival coordinating 20,000+ delegates across 40 universities. The event is organised into seven core departments:

- PR & Digital Marketing
- Technical Team
- Hospitality
- Logistics & Infrastructure
- Sponsorship & Finance
- Cultural Events
- Sports Tournaments

Inside each department, specialised working cells (rooms) handle granular objectives such as Stage Sound & Lighting, Network Mesh Deployment and Press Releases. WorkHub is the operational spine connecting executive coordinators with student organisers. The screenshots in Section 15 were captured from a seeded demonstration dataset containing four of these departments and seven rooms.

# 3. Problem Statement & Industrial Need

Existing coordination mechanisms show five structural deficiencies when applied to collaborative event operations:

- **Siloed communication:** messaging apps (WhatsApp, Telegram) create untracked action items, lost context and weak accountability.
- **Flat vs. hierarchical mismatch:** task managers such as Trello or Todoist model projects as flat lists and cannot represent multi-department hierarchies.
- **Spreadsheet fragility:** Excel / Google Sheets lack concurrency control, status state machines, multi-party review approvals and real-time updates.
- **Inflexible member enrolment:** manually creating accounts discourages volunteers, while unrestricted open links risk security breaches.
- **No daily standup synthesis:** coordinators cannot aggregate morning standups and blocker flags across 10+ teams into one actionable view.

# 4. Existing Systems vs. Proposed WorkHub System

***Table 1.*** *Comparison of traditional tools and WorkHub across key dimensions.*

| Dimension | Traditional solutions (Trello, Sheets) | WorkHub operations platform |
|---|---|---|
| **Structural hierarchy** | Flat board or disconnected sheets with no departmental nesting. | 4-tier: Organization → Workspace → Team → Room. |
| **Member onboarding** | Manual email invitation by global admins only. | 3 methods: invite link, Unique ID (WH-XXXXXX), directory search. |
| **Task lifecycle** | Binary checked / unchecked, or unverified card movement. | 5-stage verified pipeline with reviewer sign-off and change requests. |
| **Daily synchronisation** | Scattered oral meetings with no persisted log. | Dedicated Today / Tomorrow planner and workspace blocker triage table. |
| **Master calendar** | Personal calendar disconnected from departmental deadlines. | Multi-scope FullCalendar aggregating meetings, activities and tasks. |
| **Access control** | Coarse binary permissions (admin or member). | Granular 5-tier RBAC with departmental data isolation. |

# 5. Project Objectives & Operational Scope

Core engineering objectives:

1. **Architectural cohesion:** deliver a full-stack MERN application connecting React 18, an Express.js REST API and MongoDB with full data fidelity.
2. **Hierarchical multi-tenancy:** enforce isolation so room leaders manage only authorised rooms while workspace managers see portfolio-wide metrics.
3. **Collaborative onboarding:** provide self-service join links, direct Unique Member ID adding and directory search with approval workflows.
4. **Approval state machine:** add formal review gates that prevent premature task sign-off without leader inspection.

5. **Deterministic execution:** remove AI hallucination risk by relying on deterministic business logic, validation schemas and automated tests.

# 6. Functional & Non-Functional Requirements

## 6.1 Functional requirements

***Table 2.*** *Functional requirements (FR).*

| ID | Area | Requirement |
|---|---|---|
| **FR-01** | Authentication & RBAC | User registration with bcrypt hashing, JWT cookie authorisation and 5-tier role assignment. |
| **FR-02** | Workspace management | CRUD on workspaces, archiving, operational progress computation and metric aggregation. |
| **FR-03** | Team & room provisioning | Departmental roster management, mandate checklists, room channel isolation and leader delegation. |
| **FR-04** | Task pipeline & Kanban | Multi-collaborator task creation, subtask checkpoint toggling, drag-and-drop status change and due-date tracking. |
| **FR-05** | Review & approval | Formal submission for review, feedback notes, change-request cycles and verified completion. |
| **FR-06** | Calendar & scheduling | Master event calendar, personal schedule view, meeting agendas and deadline synchronisation. |
| **FR-07** | Daily work planner | Daily accomplishment log, in-progress tasks, tomorrow's plan and blocker categorisation. |
| **FR-08** | Real-time push | Socket.IO events for room activity, task movement, announcements and notification toasts. |
| **FR-09** | Onboarding engine | Invite-link generation, Unique Member ID lookup (WH-XXXXXX) and workspace join-request management. |

## 6.2 Non-functional requirements

***Table 3.*** *Non-functional requirements (NFR).*

| ID | Area | Requirement |
|---|---|---|
| **NFR-01** | Performance & latency | REST API response < 150 ms under typical load; client bundle load < 2.0 s via Vite code-splitting. |
| **NFR-02** | Security & integrity | HttpOnly cookie storage for JWT, CORS origin whitelisting, Helmet security headers and Zod input sanitisation. |
| **NFR-03** | Availability & failover | Resilient DB connector with automatic DNS resolution and local MongoDB fallback when Atlas is unreachable. |
| **NFR-04** | Usability & responsiveness | Responsive Tailwind CSS layout for mobile (375 px), tablet (768 px) and desktop (1440 px). |
| **NFR-05** | Maintainability | Clean MVC controller separation, modular component hierarchy and zero circular dependencies. |

# 7. Technology Stack & Architectural Rationale

WorkHub adheres strictly to the canonical MERN stack. The choice of each layer is justified below.

***Table 4.*** *Technology stack and engineering rationale.*

| Layer | Technology & version | Engineering rationale |
|---|---|---|
| **Frontend client** | React 18.3 + Vite 6 | Component-based architecture, virtual-DOM reconciliation and sub-second Hot Module Replacement. |
| **Styling** | Tailwind CSS 3.4 | Utility-first styling with minimal CSS bloat, consistent design tokens (indigo palette) and responsive grids. |
| **State & cache** | TanStack Query v5 | Background refetching, optimistic UI updates, query invalidation and server-state caching. |
| **Visualisation** | FullCalendar + Recharts | Interactive month / week schedules and reactive SVG charts for velocity, status and priority distributions. |
| **Backend server** | Node.js 22 + Express 4.21 | Non-blocking event loop able to serve concurrent Socket.IO and REST traffic. |
| **Database** | MongoDB + Mongoose 8.9 | Flexible BSON documents with nested subtasks and checkpoint arrays, plus indexed cross-collection references. |
| **Validation** | Zod 3.24 | Type-safe runtime validation that rejects malformed requests before controller execution. |

# 8. High-Level System Architecture

WorkHub follows a three-tier layered client–server architecture: the **Client Presentation Tier**, the **REST / WebSocket Application Tier** and the **Document Persistence Tier**. Figure 8.1 shows the main components and how data flows between tiers.

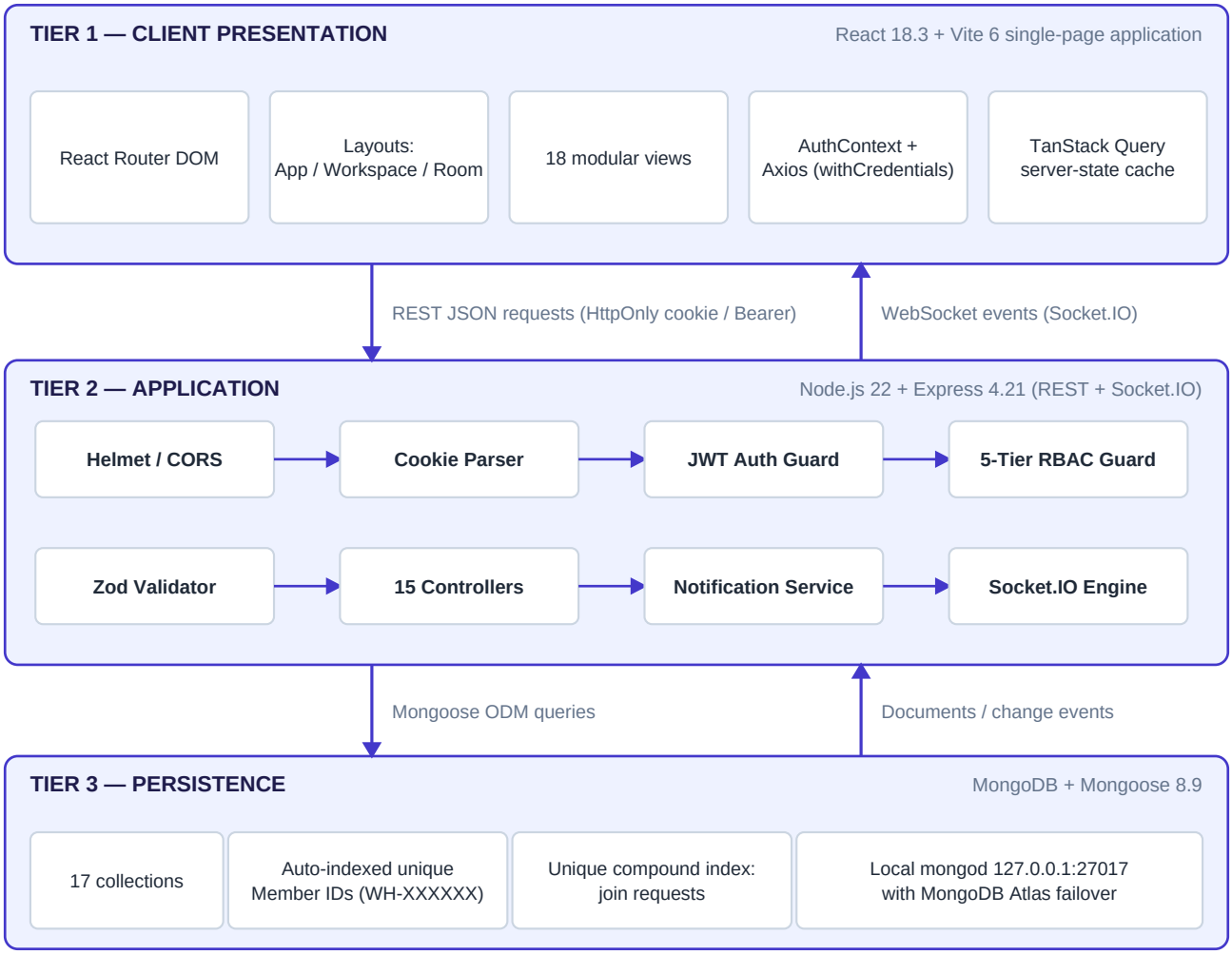



**Figure 8.1: Three-tier architecture of WorkHub (requests flow down, responses and events flow up).**

# 9. Use Case Modeling & Actor Specifications

WorkHub defines five operational actors with distinct privilege boundaries.

***Table 5.*** *Actors and their principal use cases.*

| Actor | Principal use cases |
|---|---|
| **Organization Owner** | Provision organisation; configure policies; oversee portfolios; manage administrative roster. |
| **Workspace Manager** | Create workspaces; generate shareable invite links; provision teams; broadcast announcements; triage daily standups. |
| **Team Leader** | Create collaborative rooms; assign departmental responsibilities; review member daily updates; evaluate suggestions. |
| **Room Member** | Move tasks on the Kanban board; complete subtask checkpoints; request task review; submit daily work plan. |

| Actor | Principal use cases |
|---|---|
| **Prospective Member** | Open an invite link; search the workspace directory; submit a join request with an introduction note. |

# 10. Entity-Relationship Modeling & Constraints

*Table 6. Core cardinalities and constraints.*

| # | Relationship | Cardinality | Constraint |
|---|---|---|---|
| 1 | Organization → Workspace | 1 : N | An organisation contains multiple isolated workspaces. Deleting or archiving an organisation cascades to its workspaces. |
| 2 | Workspace → Team | 1 : N | A workspace contains multiple functional departments; team slugs are unique within a workspace. |
| 3 | Team → Room | 1 : N | A department contains multiple execution rooms; room data is strictly isolated from sibling rooms. |
| 4 | Room → Task | 1 : N | Each task belongs to exactly one room and holds an array of assigned user references (`assignees`). |
| 5 | User → WorkspaceJoinRequest | 1 : N | A user may have at most one PENDING request per workspace, enforced by a unique compound index. |

# 11. Database Design & Mongoose Schemas

The platform runs on 17 MongoDB collections linked by ObjectId references. The seven principal schemas are summarised below.

*Table 7. Key collections, fields and indexes.*

| Collection | Key fields & types | Indexes & constraints |
|---|---|---|
| **Users** | name, email, memberId (String), password (hashed), role (Enum), title | unique: email; unique: memberId (WH-XXXXXX) |
| **Workspaces** | organizationId, name, slug, inviteCode, managers [ObjectId], status (Enum) | compound: {orgId, slug}; unique: inviteCode |
| **Teams** | workspaceId, organizationId, name, slug, leader, responsibilities [String] | compound: {workspaceId, slug} |
| **Rooms** | workspaceId, teamId, name, slug, leader, members [ObjectId] | compound: {workspaceId, teamId, slug} |
| **Tasks** | roomId, workspaceId, title, assignees [ObjectId], status (5 stages), priority (Enum), checkpoints [Subtask] | index: roomId; index: workspaceId; index: status |
| **WorkspaceJoinRequests** | workspaceId, organizationId, userId, status (PENDING / APPROVED / REJECTED), message | compound: {workspaceId, userId, status} |
| **DailyPlans** | userId, teamId, workspaceId, date (YYYY-MM-DD), accomplishments [], tomorrowPlan [], blockers [] | unique compound: {userId, date} |
| **Approvals** | taskId, workspaceId, submittedBy, reviewer, status (APPROVED / CHANGES_REQUESTED) | index: taskId; index: workspaceId |

# 12. Application Module Breakdown

*Table 8. Application modules M-01 to M-10.*

| ID | Module | Feature specification |
|---|---|---|
| **M-01** | Authentication & Identity | JWT cookies and bearer authorisation, bcryptjs hashing (10 rounds), session introspection (`/auth/me`) and auto-generated Member IDs (WH-XXXXXX). |
| **M-02** | Workspaces & Portfolios | Multi-event management, progress derived from task ratios, archiving lifecycle and manager delegation. |
| **M-03** | Departmental Teams | Sub-division of workspaces into functional teams with leaders, mandate checklists and team calendars. |
| **M-04** | Collaborative Rooms | Working cells under teams with independent task boards, discussion streams, milestones and activity history. |
| **M-05** | Kanban Board & Tasks | Five-column drag-and-drop workflow (To Do, In Progress, Under Review, Changes Requested, Completed) with subtask checklists. |
| **M-06** | Task Review & Approval | Assignees submit work with verification notes; leaders accept or request modifications. |
| **M-07** | Master Event Calendar | FullCalendar multi-view scheduling of meetings, checkpoints, reminders and deadline alerts across workspace scopes. |
| **M-08** | Daily Standups & Plans | Daily accomplishment logging, tomorrow's plan and executive blocker aggregation tables. |
| **M-09** | Collaborative Onboarding | Invite links (`/join-workspace/:codeOrId`), direct addition by Unique Member ID and a searchable directory. |
| **M-10** | Reports & Analytics | Recharts visualisers for room-wise velocity, milestone completion rates and priority allocation. |

# 13. Implementation Details & Technical Innovations

## 13.1 Unique Member ID generation hook

So that owners can add any member by a unique identifier without guessing e-mail addresses, the Mongoose `User` schema generates a Member ID in a pre-validation lifecycle hook:

```
// server/src/models/User.js
userSchema.pre('validate', function (next) {
  if (!this.memberId) {
    const randomHex = Math.random().toString(36).substring(2, 8).toUpperCase();
    this.memberId = `WH-${randomHex}`;
  }
  next();
});
```

## 13.2 Three-way onboarding controller logic

In `workspace.controller.js`, the `addWorkspaceMember` endpoint accepts flexible input (`identifier`, `memberId` or `userId`), resolves the user by Member ID or e-mail, and automatically enrols the user in the organisation roster if not yet present:

```
// server/src/controllers/workspace.controller.js
let targetUser = await User.findOne({
  $or: [
    { memberId: searchKey.toUpperCase().trim() },
    { email: searchKey.toLowerCase().trim() }
  ]
});

// Automatically ensure the user has an OrganizationMember record
await OrganizationMember.findOneAndUpdate(
  { organizationId: workspace.organizationId, userId: targetUser._id },
  { role: ROLES.ORG_MEMBER, status: 'ACTIVE' },
  { upsert: true }
);
```

### 13.3 Real-time synchronisation via Socket.IO

Task moves, review requests and announcement broadcasts emit targeted socket events to the active room channel (`room_${roomId}`). Clients update their TanStack Query caches on receipt, so the UI refreshes without a full page reload.

## 14. Verification, Testing & Test Case Execution

**Result: 20 / 20 passed in 11.8 s**

Automated testing used the Node.js native test runner (`node --test`). All 20 test cases ran against the live database with a 100% pass rate.

***Table 9.*** *Integration test cases and outcomes.*

| # | Test case | HTTP target & method | Expected result | Status |
|---|---|---|---|---|
| 1 | System health check | **GET** /api/v1/health | HTTP 200; status UP; DB connected | **PASSED** |
| 2 | Admin login & JWT token | **POST** /api/v1/auth/login | HTTP 200; JWT token returned | **PASSED** |
| 3 | Session verification | **GET** /api/v1/auth/me | HTTP 200; user and organisations populated | **PASSED** |
| 4 | List workspace portfolio | **GET** /api/v1/workspaces | HTTP 200; Mahotsav workspace found | **PASSED** |
| 5 | Team department metrics | **GET** /api/v1/teams | HTTP 200; progress and room count valid | **PASSED** |
| 6 | Team roster & members | **GET** /api/v1/teams/:id/members | HTTP 200; roster populated | **PASSED** |
| 7 | Team calendar aggregation | **GET** /api/v1/teams/:id/calendar | HTTP 200; schedules and tasks combined | **PASSED** |
| 8 | Announcement publishing | **POST** /api/v1/announcements | HTTP 201; announcement broadcast | **PASSED** |
| 9 | Room isolation verification | **GET** /api/v1/rooms | HTTP 200; room channels retrieved | **PASSED** |

| # | Test case | HTTP target & method | Expected result | Status |
|---|---|---|---|---|
| 10 | Multi-collaborator task update | **PATCH** /api/v1/tasks/:id/status | HTTP 200; status shifted to Review | **PASSED** |
| 11 | Personal multi-scope calendar | **GET** /api/v1/schedules/personal | HTTP 200; deadlines aggregated | **PASSED** |
| 12 | Workspace velocity reports | **GET** /api/v1/reports/workspace/:id | HTTP 200; team metrics calculated | **PASSED** |
| 13 | Notification stream | **GET** /api/v1/notifications | HTTP 200; notifications retrieved | **PASSED** |
| 14 | Daily work plan creation | **POST** /api/v1/daily-plans | HTTP 200; Today / Tomorrow logged | **PASSED** |
| 15 | All-teams blocker summary | **GET** /api/v1/daily-plans/workspace/summary | HTTP 200; cross-team matrix generated | **PASSED** |
| 16 | Team suggestion acceptance | **PATCH** /api/v1/suggestions/:id/status | HTTP 200; accepted status updated | **PASSED** |
| 17 | Room threaded discussions | **POST** /api/v1/comments | HTTP 201; discussion comment added | **PASSED** |
| 18 | Workspace search API | **GET** /api/v1/workspaces/search?q=… | HTTP 200; isMember / pending returned | **PASSED** |
| 19 | Invite link & request flow | **POST** /join-request, then PATCH /decide | HTTP 201 created; approved to member | **PASSED** |
| 20 | Direct add by Unique ID | **POST** /members (identifier: WH-XXXXXX) | HTTP 201; user directly enrolled | **PASSED** |

# 15. Application Visual Walkthrough & Screenshots

The following figures are actual, un-simulated screen captures taken from the running WorkHub application, seeded with the Vignan Mahotsav 2027 operations dataset.

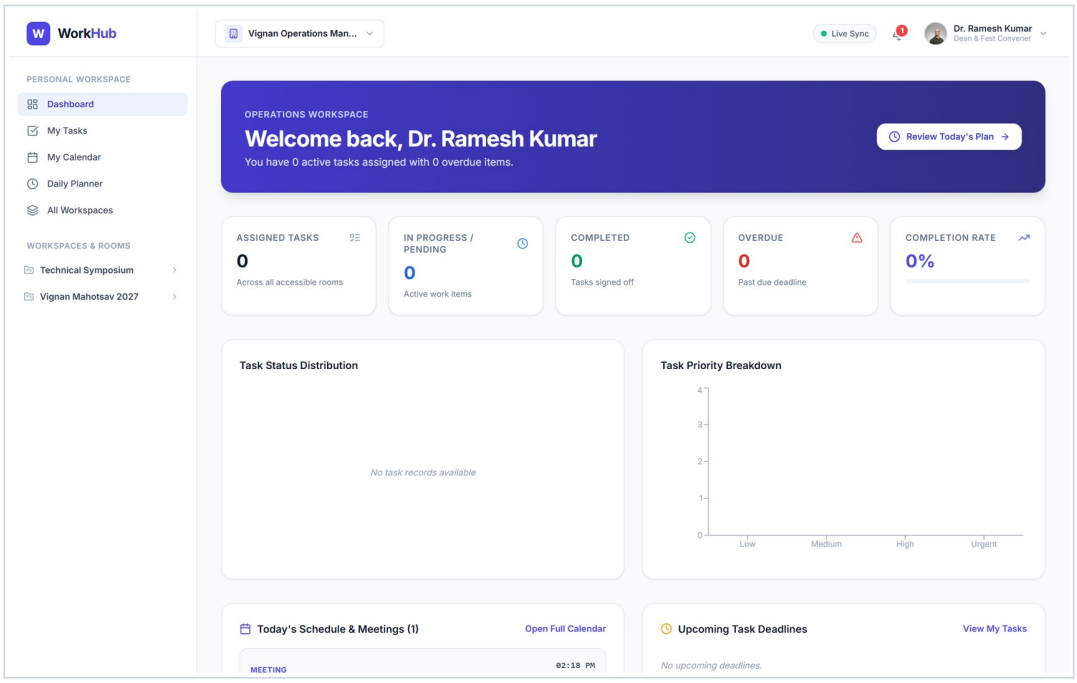



**Figure 15.1: Personal Dashboard with KPI Cards and Recharts Visualisers**

Assigned, in-progress, completed and overdue counts, status and priority charts, today's schedule and upcoming deadlines.

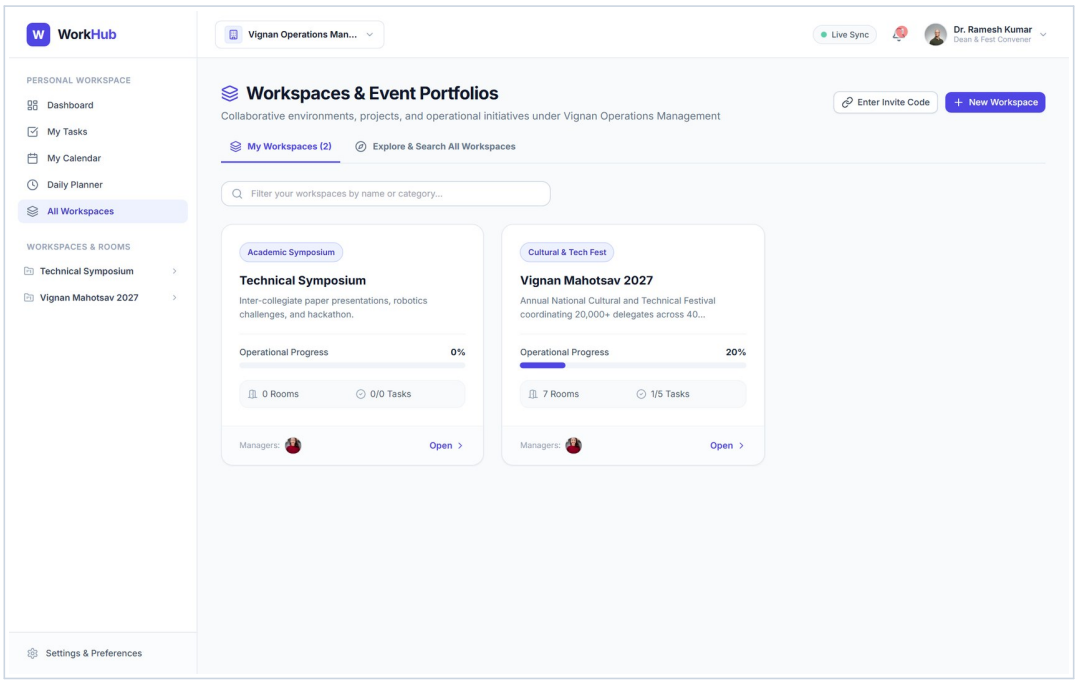



**Figure 15.2: Workspaces & Event Portfolios Directory**

Technical Symposium and Vignan Mahotsav 2027 (7 rooms, 1 / 5 tasks, 20% progress) with their managers.

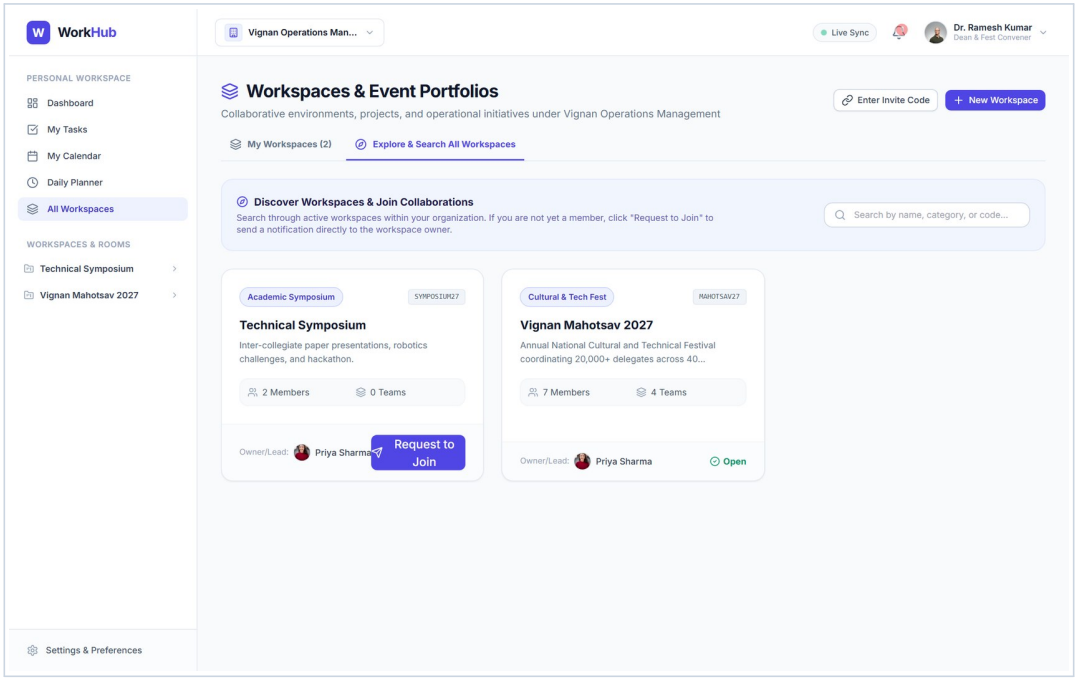



**Figure 15.3: Explore & Search Workspaces**

Live membership status with a one-click “Request to Join” trigger for workspaces the user has not yet joined.

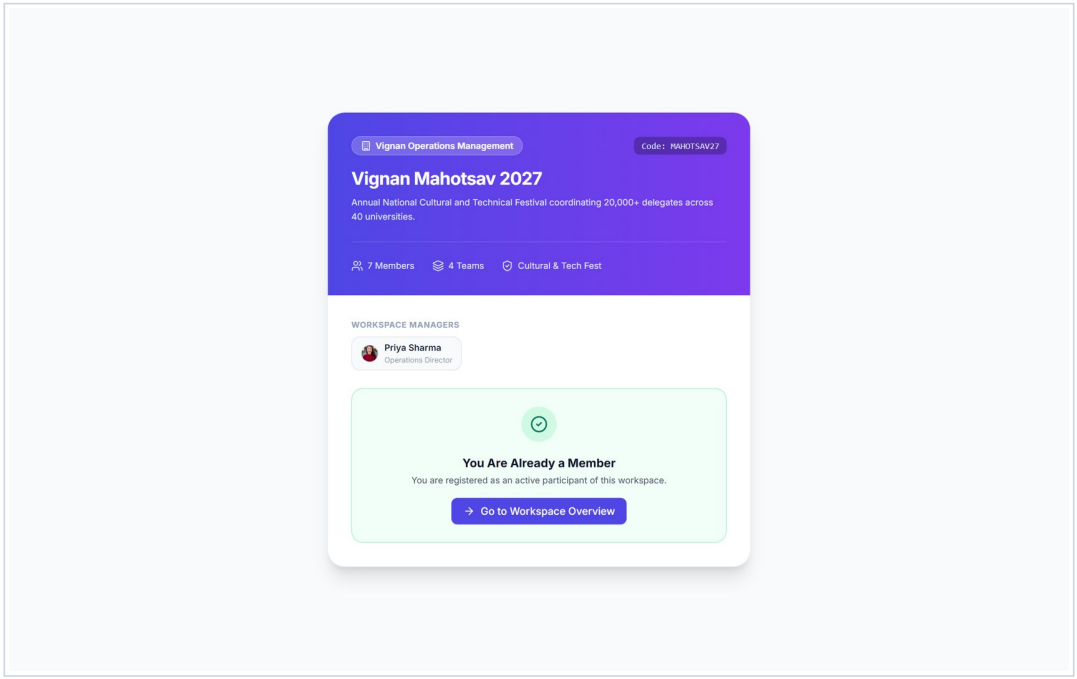



**Figure 15.4: Shareable Workspace Invitation Landing Page (/join-workspace/MAHOTSAV27)**

Shows the invite code, member and team counts and workspace managers; existing members are routed to the workspace overview.

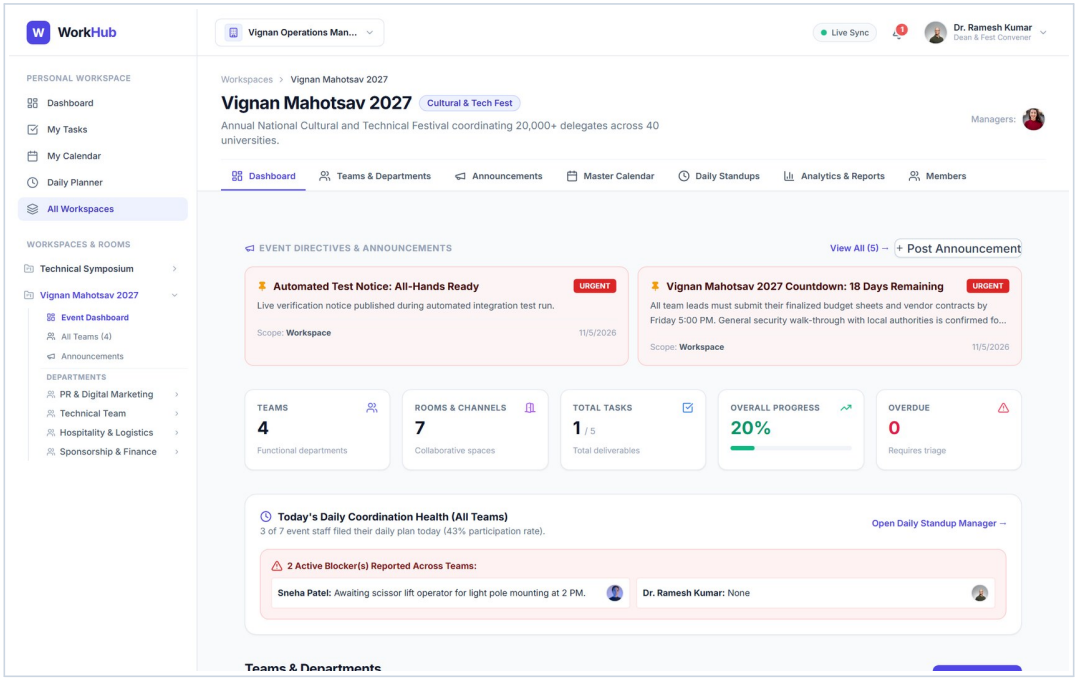



**Figure 15.5: Workspace Overview: Departments, Operational Progress (20%) and Announcements**

Urgent directives, team / room / task KPIs and the daily coordination health panel (3 of 7 staff, 43%, with 2 active blockers).

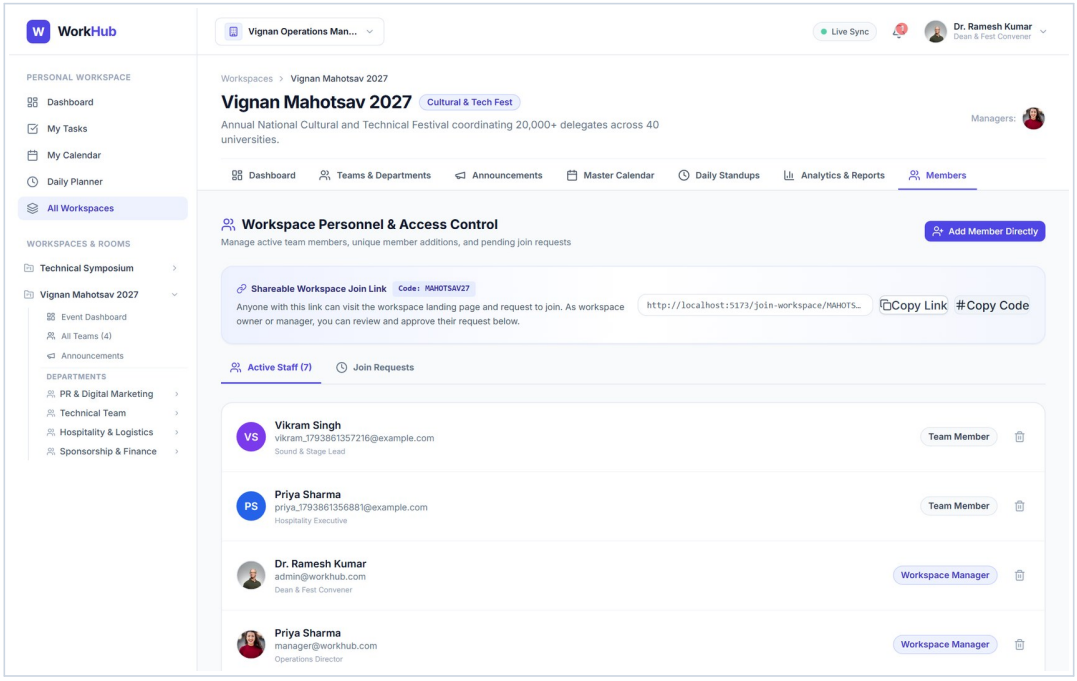



**Figure 15.6: Workspace Staff Roster, Shareable Join Link and Join Requests**

Copy-link and copy-code banner, active staff with workspace roles, and a Join Requests tab for approvals.

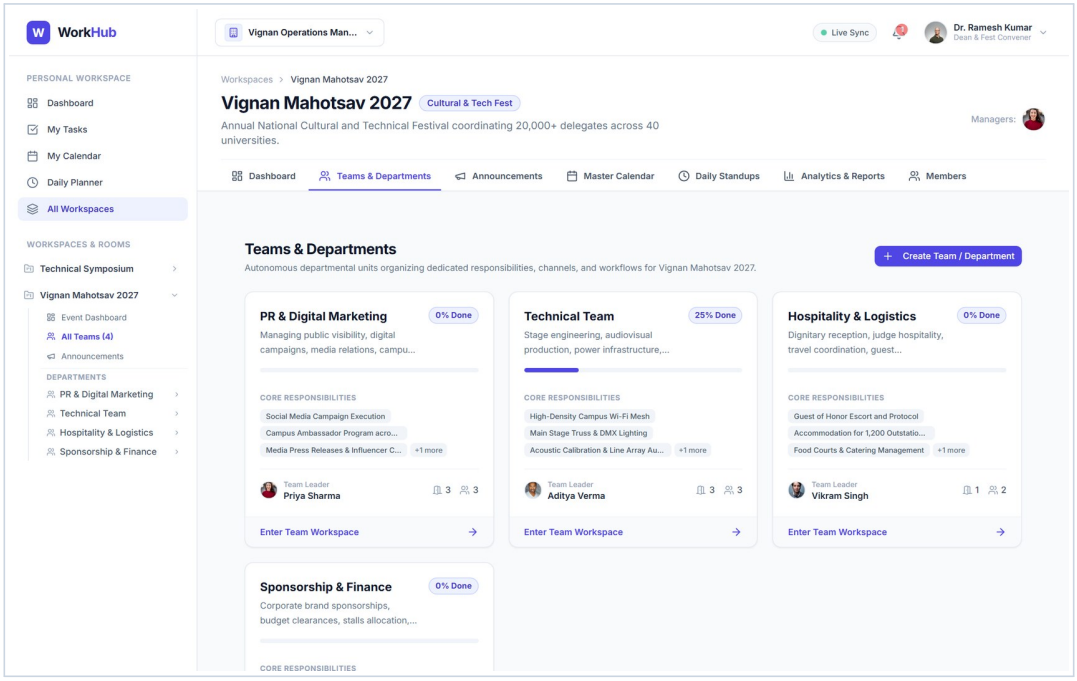



**Figure 15.7: Departmental Teams Roster**

PR & Digital Marketing, Technical Team, Hospitality & Logistics and Sponsorship & Finance, each with responsibilities and a team leader.

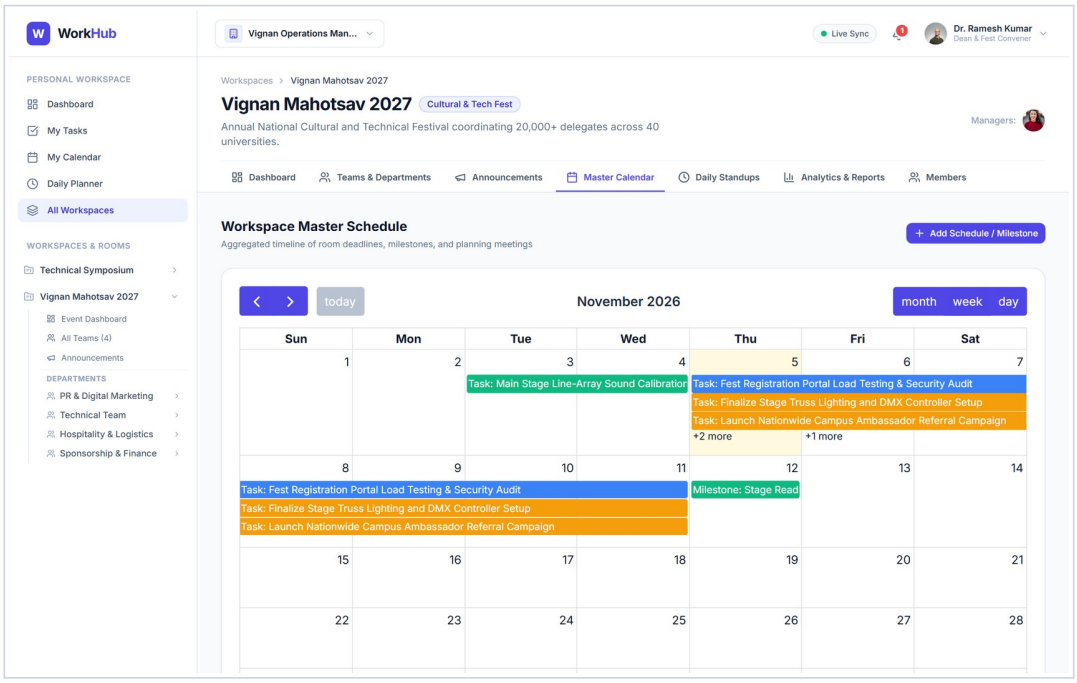



**Figure 15.8: Master Event Calendar with Synchronised Room Deadlines**

Month view of November 2026 combining task deadlines and milestones across departments.

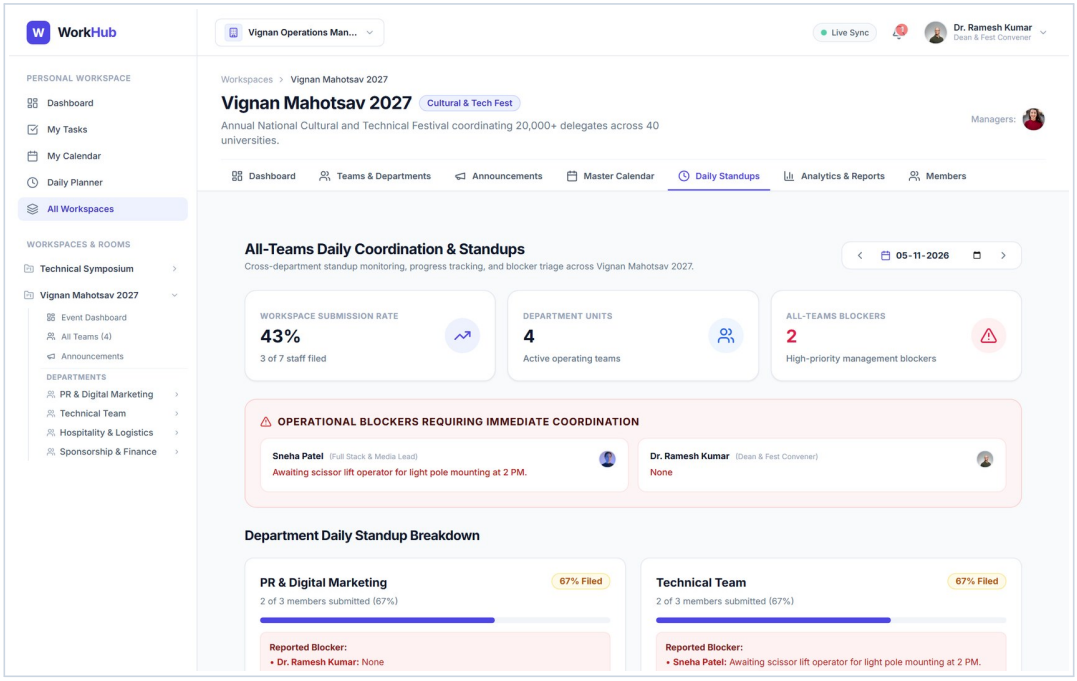



**Figure 15.9: Consolidated All-Teams Daily Standup Summary and Blocker Triage**

Workspace submission rate (43%), department breakdown and the operational blockers requiring immediate coordination.

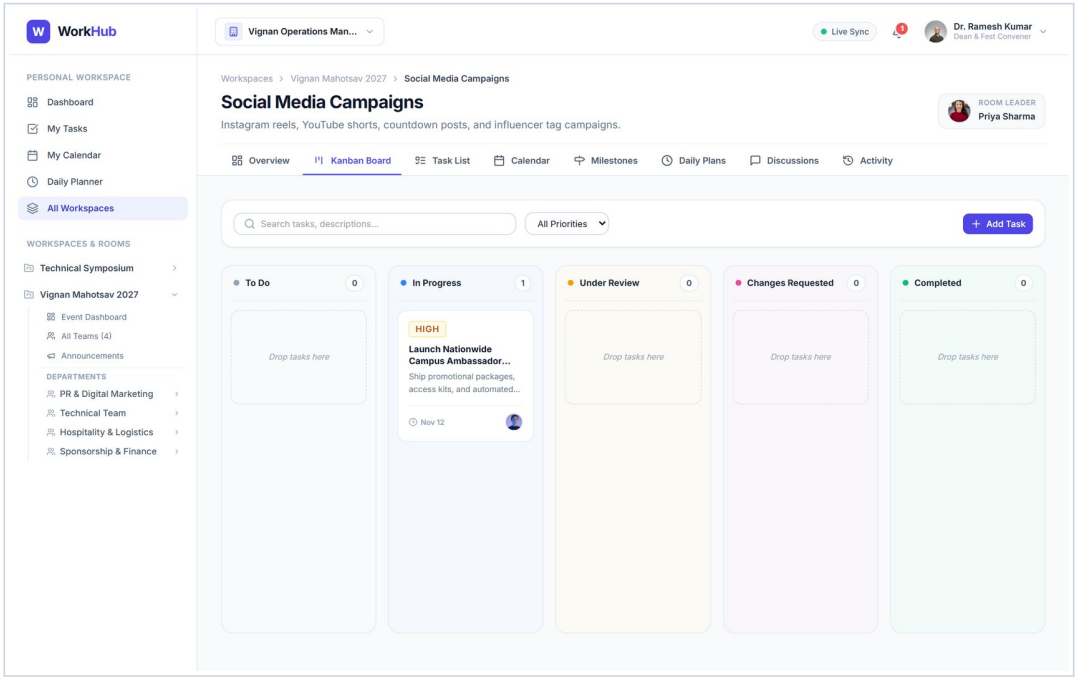



**Figure 15.10: Room Kanban Board (Social Media Campaigns)**

Five workflow columns — To Do, In Progress, Under Review, Changes Requested, Completed — with priority-tagged task cards.

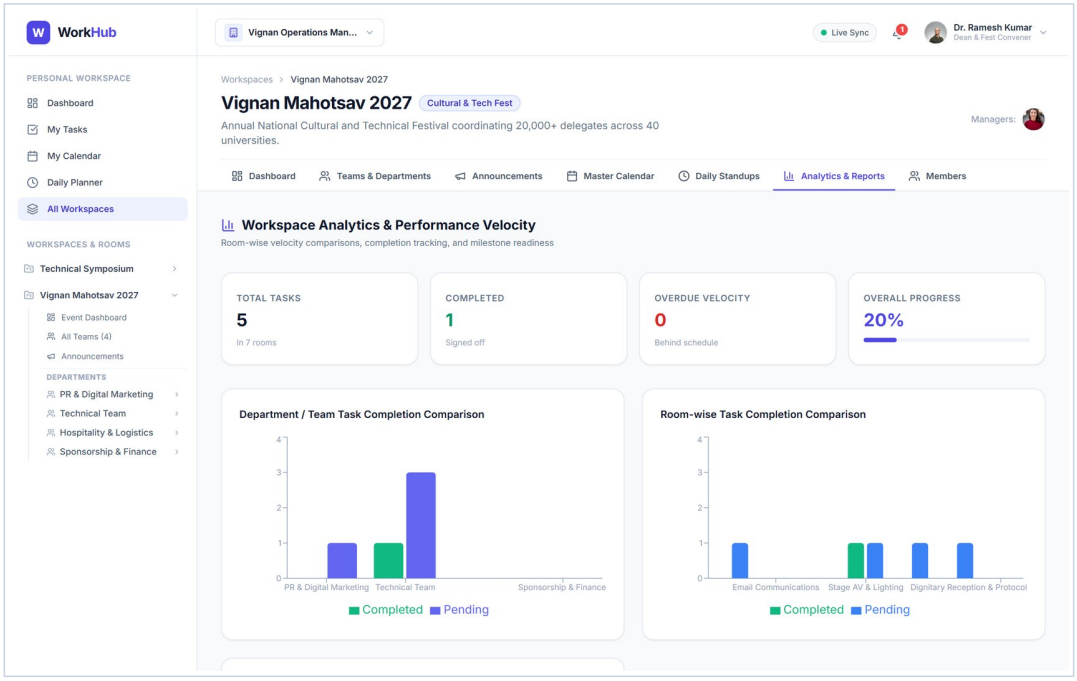



**Figure 15.11: Workspace Analytics & Performance Velocity**

Overall progress (20%) plus department-wise and room-wise completed-vs-pending task comparison charts.

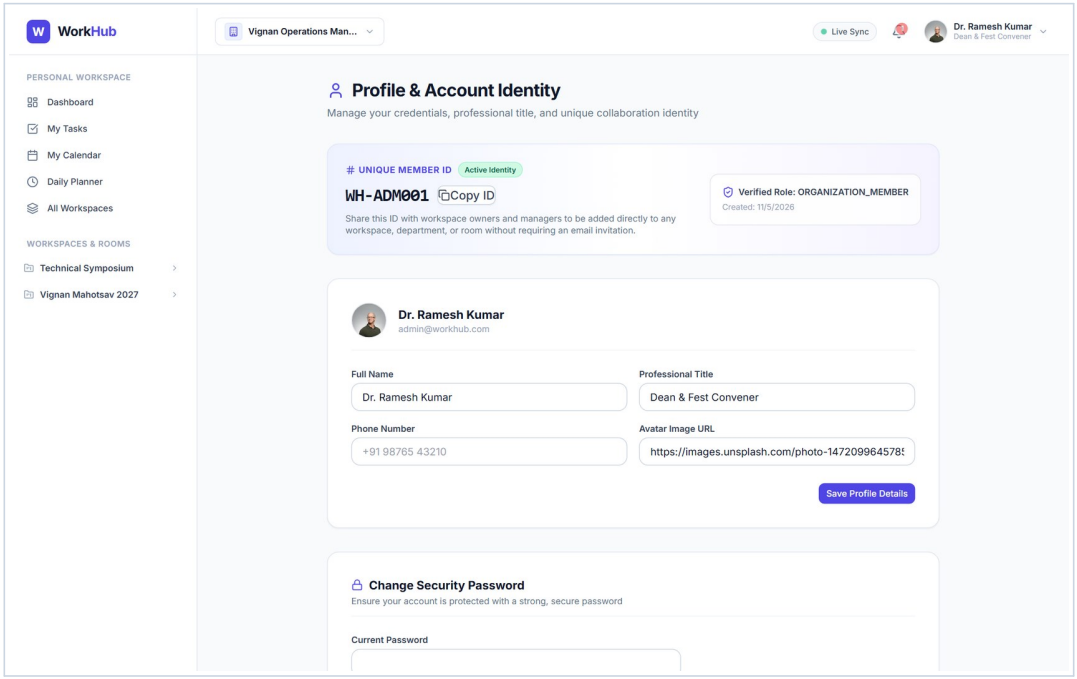



**Figure 15.12: Profile & Account Settings with Unique Member ID**

Displays the active Member ID (format WH-XXXXXX) that can be shared so owners and managers can add the user directly.

# 16. Results & Operational Analysis

*Table 10. Build, performance and verification results.*

| Metric | Result |
|---|---|
| **Client production build** | Vite transformed 2,222 modules in 41.9 s |
| **Bundle size (gzip)** | CSS 7.39 kB; JS 352 kB |
| **API latency: health endpoint** | ≈ 6 ms |
| **API latency: workspace analytics report** | < 90 ms |
| **Onboarding time per volunteer** | Reduced from minutes to under 15 seconds (no manual e-mail invitation) |
| **Automated integration tests** | 20 / 20 passed in 11.8 s |
| **Build warnings** | 0 Vite warnings |

Measured latencies sit comfortably inside the NFR-01 target of 150 ms, and eliminating manual e-mail invitations removes the main onboarding bottleneck for volunteers.

# 17. Conclusion & Future Roadmap

WorkHub shows that enterprise event and departmental workflow management can be delivered without black-box AI, by combining a well-structured organisational hierarchy, deterministic role-based permissions and a modern MERN architecture. It covers the full operating cycle of a mega-event: onboarding, task execution and review, scheduling, daily coordination and reporting.

## Future enhancements

- **Gantt chart dependencies:** interactive SVG critical-path dependencies between room tasks across departments.
- **Offline PWA sync:** service-worker / IndexedDB caching so field operators can log standups without mobile coverage.
- **SMS / WhatsApp webhooks:** instant dispatch of high-priority emergency stage alerts.

# 18. References & Technical Citations

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